Inversion for the G3c logical rules under closed-surface cut, preserving height and cut structure

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Editorial note. Scope declaration: the author demonstrates the cut case for one rule (left-and) with both derivations displayed and closes with the two invariants; the remaining logical rules are instances of the same shape, which this card declares rather than re-derives. No author-marked omission occurs inside the counted span.

Editorial note. The counted statement excludes right-exists and left-forall, the two rules whose inversion fails in G3c; the counted argument does not rely on the deferred second half of the quantifier inversion lemma.

Editorial note. The two cross-unit references in the proof's opening sentence point at the sequent-calculus inversion lemmas whose case structure the counted proof follows and restates; both are declared as navigational references.

Editorial scope. One original lemma of the unit's cut-elimination section, counted once: inversion for the logical rules of G3c with closed-surface cut, with height and cut-structure invariants. The counted lemma and its author proof are retained in full; the surrounding discussion of cut rank before the statement and the remark after the proof are not counted.

Allowed standard prerequisites (editorial).

- The one-sided classical sequent calculus G3c extended by the closed-surface cut rule, the height of a proof, and the cut rank of a cut inference, defined as the depth of its cut formula.
- The height-preserving inversion lemma for the propositional rules of G3c and the inversion lemma for the quantifier rules of G3c, both proved in the sequent-calculus unit of the same source tree; the counted proof follows their case structure, which it restates in full.
- The induction-with-varying-contexts convention: in height-preserving arguments the inductive hypothesis is stated for all contexts, not just the ones occurring in the given proof.

Source reference. Original lemma G3c-invert: sources/olp-proof-theory-sequent-calculus/pinned-original-source.tex, source lines 963–963. The counted proof

follows the case structure of the height-preserving inversion lemma for the G3c propositional rules proved in the sequent-calculus unit; the case walk is restated inside the counted proof. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. Published as corpus card X2-proof-theory-sequent-calculus-g3c-invert.

Source reference. Original lemma invert-quant: sources/olp-proof-theory-sequent-calculus/pinned-original-source.tex, source lines 1081–1081. The counted proof follows the case structure of the quantifier inversion lemma proved in the sequent-calculus unit; the counted statement excludes exactly the two rules whose inversion fails, and the case walk is restated inside the counted proof. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full.

Lemma 0.1. *If* **G3c** + Cut *proves the conclusion of a logical rule R other than or using proof π then it also proves each premise with proof π' (or proofs π', π'' depending on whether the rule has one or two premises). Moreover, (a)* $\operatorname{ht}\pi' \leq \operatorname{ht}\pi$ *(and* $\operatorname{ht}\pi'' \leq \operatorname{ht}\pi$*) and (b) the number of Cut inferences as well as their ranks is the same in π' (and π'') as in π.*

Proof. By inspection of the proofs of original lemmas G3c-invert and invert-quant. That proof was by induction on $\operatorname{ht}\pi$. In the base case, we replaced an axiom by another axiom. So conditions (a) and (b) are trivially satisfied. If the last rule of π was R, its immediate sub-proofs are the proofs π_i we want, and (a) and (b) are both satisfied. In all other cases, we applied the inductive hypothesis to the immediate sub-proofs of π, i.e., those leading to the premise(s) of the last inference R, and then applied that same rule R to the results. The conditions (a) and (b) hold for the immediate sub-proofs of the new proof, and so (a) and (b) hold for the entire proof. It remains to verify the case where the last inference is Cut. We do this again for one example only: suppose π is proof of the conclusion $B \wedge C, \Gamma \Rightarrow \Delta$ of , and ends in Cut:

$$\dfrac{\begin{array}{cc}\vdots\,\pi_1 & \vdots\,\pi_2 \\ B \wedge C, \Gamma\Delta, A & A, B \wedge C, \Gamma\Delta\end{array}}{B \wedge C, \Gamma\Delta}\ \text{Cut}$$

The induction hypothesis applies to π_1 and π_2 (which are also proofs of instances of the conclusion of) and yields proofs π_1' and π_2' of $B, C, \Gamma\Delta, A$ and $A, B, C, \Gamma\Delta$, respectively. We combine these using Cut to obtain π':

$$\dfrac{\begin{array}{cc}\vdots\,\pi_1' & \vdots\,\pi_2' \\ B, C, \Gamma\Delta, A & A, B, C, \Gamma\Delta\end{array}}{B, C, \Gamma\Delta}\ \text{Cut}$$

Clearly, (a) and (b) are satisfied. □